
SOFTWARE REQUIREMENTS SPECIFICATION

for the

MATA – Multimodal Accessible Textbook Assistant (Mobile Application)

AI-Powered · Camera-First · Offline-Ready · Accessibility-Focused

Prepared in conformance with IEEE Std 830-1998

IEEE Recommended Practice for Software Requirements Specifications

Prepared by:

Student Name	Exam Seat No.	Role
Ruturaj Kasture	202301040275	Team Lead / AI Pipeline
Maitreyee Majumdar	202301040277	Backend / LLM Integration
Kanishka Amrutkar	202301040244	Mobile App / UI
Amisha Singh	202301100041	OCR / Vision Module

Project Guide: Mrs. Neha Hajare

Project Coordinator: Dr. Kanchan Dhote | HoD: Dr. Pramod Ganjewar

Department of Computer Engineering

MIT Academy of Engineering, Alandi (D), Pune – 412105

(Affiliated to Savitribai Phule Pune University)

Document Version: 2.0 (Application Edition) | April 2026

Revision History

Version	Date	Author(s)	Change Description
1.0	Dec 2025	All Members	Initial SRS – Web-based system
2.0	Apr 2026	All Members	Revised SRS – Converted to Mobile Application with new market features
2.1	TBD	Dev Team	Post-Sem 2 updates after implementation
3.0	TBD	Dev Team	Final SRS after Sem 3 deployment

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for MATA (Multimodal Accessible Textbook Assistant) — a native mobile application targeting Android and iOS platforms. This version supersedes the earlier web-based SRS (v1.0) and incorporates the paradigm shift from a browser-hosted system to a camera-first, device-native mobile application.

MATA is an AI-powered educational accessibility tool that allows students to point their phone camera at any textbook page — or upload an image/PDF — and instantly receive a spoken, age-adapted explanation of all content including text, diagrams, charts, mathematical expressions, and structured layouts. The application is designed to serve visually impaired students, young learners (ages 8–16), and anyone who benefits from audio-based learning.

This document is prepared in conformance with IEEE Std 830-1998 and is intended for: the development team, project guide (Mrs. Neha Hajare), project coordinator (Dr. Kanchan Dhote), external evaluators, and future maintainers.

1.2 Document Conventions

Convention	Meaning
SHALL	Mandatory requirement – must be implemented.
SHOULD	Recommended – strongly desired but not mandatory in MVP.
MAY	Optional – implement if time/resources allow.
FR-XX	Functional Requirement identifier (e.g., FR-01).
NFR-XX	Non-Functional Requirement identifier.
UC-XX	Use Case identifier.
MVP	Minimum Viable Product – core features for Semester 2 demo.
[APP]	Requirement specific to mobile app context (replaces web equivalent).

1.3 Intended Audience

Audience	Relevant Sections
Development Team	All sections, especially §3 (Functional Req.) and §4 (App Architecture)
Project Guide / Coordinator	§1, §2, §3, §5, §6
External Examiners	§1, §2, §3.1–§3.3, §6
Future Maintainers	§3, §4, §5, Appendices

Audience	Relevant Sections
Investors / Market Stakeholders	§2, §3.1, §7 (Unique Market Features)

1.4 Project Scope

MATA — as a mobile application — provides the following core scope:

- Camera-first textbook scanning: Users open the app and point the camera at any textbook page for instant real-time capture.
- Full multimodal AI pipeline running on-device (lightweight models) or via cloud inference (larger models): OCR → Layout Analysis → Vision Understanding → LLM Explanation → TTS Output.
- Three explanation modes: Child (8–12), Teen (13–16), Adult/Standard.
- Audio playback of explanations with play/pause/speed controls.
- Interactive Q&A: users type or voice-input follow-up questions.
- Offline-capable MVP: Core OCR and TTS to work without internet; LLM via cloud API with graceful degradation.
- Accessibility-first UI: Screen-reader compatible, high-contrast mode, large tap targets.

Out of Scope (v2.0):

- Full LMS (Learning Management System) integration.
- Teacher dashboard / classroom management features.
- Real-time AR overlay on camera feed (planned for v3.0).
- Full multilingual support beyond English (Semester 3 roadmap).

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
MATA	Multimodal Accessible Textbook Assistant – the mobile application.
OCR	Optical Character Recognition – text extraction from images.
LLM	Large Language Model – AI model for natural language generation.
TTS	Text-to-Speech – converts text to audio output.
STT	Speech-to-Text – converts voice input to text (for voice Q&A).
ViT	Vision Transformer – image feature extraction model.
DETR	Detection Transformer – object detection in images.
VLM	Vision-Language Model (e.g., LLaVA) – understands images + text.

Term	Definition
API	Application Programming Interface.
SDK	Software Development Kit.
Flutter	Google's cross-platform mobile UI framework (Dart language).
FastAPI	Python-based REST API backend framework.
MVP	Minimum Viable Product.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
SRS	Software Requirements Specification.
CER	Character Error Rate – OCR accuracy metric.
MOS	Mean Opinion Score – TTS quality metric (1–5 scale).
BLEU	Bilingual Evaluation Understudy – NLP text quality metric.
WCAG	Web Content Accessibility Guidelines (also applicable to mobile).
SPPU	Savitribai Phule Pune University.
MITAOE	MIT Academy of Engineering.

1.6 References

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
- M. Li et al., 'TrOCR: Transformer-based OCR with Pre-trained Models,' AAAI 2023.
- A. F. Biten et al., 'LaTr: Layout-Aware Transformer for Scene-Text VQA,' CVPR 2022.
- P. Yan et al., 'Context-Aware Chart Element Detection (CACHED),' ICDAR 2023.
- Y. Fu et al., 'EDSL: Encoder-Decoder for Mathematical Expression Recognition,' ICDAR 2023.
- M. Le et al., 'Voicebox: Text-Guided Multilingual Speech Generation at Scale,' NeurIPS 2023.
- M. Schorlemmer et al., 'Generating Qualitative Descriptions of Diagrams,' Diagrams 2024.
- G. Li et al., 'Aligning Sentence Simplification with ESL Learner Proficiency,' NAACL 2025.
- X. Zhou et al., 'Deep Learning Based Knowledge Tracing,' Scientific Reports 2025.
- Flutter Documentation – <https://flutter.dev/docs>
- Android Accessibility Guidelines – <https://developer.android.com/guide/topics/ui/accessibility>
- Apple HIG Accessibility – <https://developer.apple.com/design/human-interface-guidelines/accessibility>

2. Overall Description

2.1 Product Perspective

MATA is a standalone native mobile application — not a module of a larger existing system. The shift from a web-based architecture (SRS v1.0) to a mobile application is driven by the following critical insights:

Why Mobile App over Website?

1. Camera-First Access: The primary input for a student is pointing the phone at a textbook — this is natural on mobile, unnatural on a website.
2. Offline Capability: Students in rural India or low-connectivity areas cannot rely on browser-based cloud systems. A mobile app can run lightweight models offline.
3. Accessibility APIs: Android TalkBack and iOS VoiceOver provide native screen-reader integration that a website cannot match.
4. Push Notifications: App can remind students to continue learning, something impossible in a passive website.
5. App Store Distribution: Deploying via Google Play / App Store reaches millions of students directly without any setup friction.

The application communicates with a cloud backend (FastAPI server) for heavy AI inference (LLM, VLM) while performing lightweight tasks (OCR, TTS) on-device when possible. The architecture is client-heavy for speed and offline-resilience.

2.2 Product Functions

MATA's primary functions as a mobile app:

1. Camera Capture – Real-time viewfinder to photograph textbook pages.
2. Gallery / PDF Import – Upload from phone gallery or file manager.
3. On-Device Preprocessing – Image deskew, crop, enhance before sending to AI.
4. OCR + Layout Analysis – Extract and structure all text from the page.
5. Visual Content Understanding – Describe diagrams, charts, equations via VLM.
6. Age-Adaptive Explanation – LLM generates explanations tuned to Child/Teen/Adult mode.
7. Text-to-Speech Playback – Native + neural TTS with playback controls.
8. Voice / Text Q&A – Interactive follow-up via microphone or keyboard.
9. Offline Core – OCR + basic TTS functional without internet.
10. Learning History – Session log; saved explanations accessible later.
11. Adaptive Feedback – Difficulty rating adjusts next explanation complexity.
12. Accessibility Mode – High contrast, large text, screen-reader compatible.

2.3 User Classes and Characteristics

User Class	Age	Description	App Usage Pattern
Visually Impaired Student	Any	Relies entirely on TTS audio; cannot see screen clearly.	Camera capture → voice Q&A → audio playback only.
Primary Learner	8–12	Young student; needs child-mode explanations.	Parent-assisted capture → listens to explanation.
Secondary Learner	13–16	Self-sufficient; uses teen mode; may ask questions.	Self-capture → audio + text view → typed Q&A.
Adult / Self-Learner	17+	College student or adult; uses full detail mode.	PDF import → detailed explanation → research Q&A.
Teacher / Educator	25+	Demos the app in class; may use it for lesson prep.	Classroom demonstration; bulk page scan.
Parent	30+	Helps younger child use the app.	Assists child in capturing pages; monitors history.
Admin / Developer	Any	Manages backend AI models and monitors performance.	Backend dashboard (separate admin panel).

2.4 Operating Environment

2.4.1 Mobile Client Environment

Parameter	Requirement
Target Platforms	Android 10.0+ (API Level 29+) and iOS 14.0+
Development Framework	Flutter 3.x (Dart) – single codebase for Android + iOS
Minimum RAM	3 GB (for on-device OCR inference)
Storage	100 MB app install + 500 MB for offline model cache
Camera	Rear camera with autofocus, minimum 8 MP
Microphone	Required for voice Q&A (STT) feature
Speaker / Headphones	Required for TTS audio output
Connectivity	Wi-Fi / 4G for LLM+VLM cloud calls; offline mode available for OCR+TTS
Screen Size	4.5 inches minimum; optimized for 6–7 inch displays

2.4.2 Backend Server Environment

Component	Specification
OS	Ubuntu 22.04 LTS
Language / Framework	Python 3.10+ / FastAPI 0.100+
GPU	NVIDIA RTX 3060+ (12 GB VRAM) for LLM+VLM inference
RAM	32 GB DDR4

Component	Specification
Storage	100 GB SSD for model weights + logs
API Protocol	REST over HTTPS; JSON payloads; multipart/form-data for image uploads
Deployment	Docker + NGINX reverse proxy; optional: cloud (AWS/GCP/Azure)
Database	PostgreSQL 14+ for user profiles; Redis for session caching

2.5 Design and Implementation Constraints

- [APP] The mobile app MUST be built using Flutter for cross-platform (Android + iOS) deployment from a single codebase, reducing development overhead.
- [APP] On-device inference (OCR, lightweight TTS) MUST use quantized ONNX or TensorFlow Lite models to run within 3 GB RAM without draining battery.
- [APP] The app MUST NOT require root/jailbreak access to any device functionality.
- LLM and Vision-Language Model inference MUST run on the cloud backend due to model size (7B–72B parameters); results are streamed back to the app.
- [APP] Camera integration MUST comply with Android Camera2 API and iOS AVFoundation framework guidelines.
- [APP] All personal data (user profiles, session logs) stored on-device MUST be encrypted using AES-256 via Flutter Secure Storage.
- Diagram interpretation accuracy for complex illustrations remains an active research limitation; the app MUST clearly indicate confidence levels to users.
- [APP] App binary size MUST remain below 100 MB (excluding downloadable model cache) to comply with Play Store and App Store size guidelines.

2.6 Assumptions and Dependencies

2.6.1 Assumptions

- The device camera produces images of sufficient quality (≥ 8 MP) for OCR.
- Internet connectivity (Wi-Fi or 4G) is available for LLM/VLM-dependent features.
- The primary language of textbook content is English in the initial version.
- The user selects their age/difficulty level during first-time app onboarding.
- Students upload single pages, not multi-page spreads (each page processed individually).

2.6.2 Dependencies

- Flutter SDK 3.x and Dart 3.x – mobile development framework.
- Hugging Face Transformers – pre-trained AI model weights (TrOCR, LLaVA, LLaMA).

- Coqui TTS / Piper – open-source neural TTS (MIT/Mozilla license).
- Google MLKit Text Recognition – on-device OCR fallback for Android.
- Apple Vision Framework – on-device OCR fallback for iOS.
- FastAPI + Uvicorn – backend REST API server.
- PostgreSQL + Redis – backend data persistence and caching.

3. Specific Requirements

This section is the primary contractual core of the SRS. All requirements use the priority classification: High (MVP – must ship), Medium (Semester 3), Low (Future scope).

3.1 External Interface Requirements

3.1.1 User Interface – Mobile App [APP]

ID	Requirement	Priority
FR-01	The app SHALL have a bottom navigation bar with tabs: Home (Scan), History, Profile, Settings.	High
FR-02	The Home screen SHALL display a camera viewfinder occupying at least 70% of the screen with a prominent Capture button.	High
FR-03	The app SHALL provide a Gallery Import button to select images from the device photo library.	High
FR-04	The app SHALL provide a PDF Import option via the device file manager.	High
FR-05	After processing, results SHALL be displayed on a Result Screen with: explanation text (scrollable), audio player bar (play/pause/speed), and a Q&A input field.	High
FR-06	The app SHALL include an Age Level selector (Child / Teen / Adult) accessible from the Result Screen and from Settings.	High
FR-07	The app SHALL support a High-Contrast Accessibility Mode toggled from Settings.	High
FR-08	The app SHALL support system font-size scaling (small / medium / large / extra-large).	Medium
FR-09	All interactive elements SHALL have minimum tap target size of 48x48 dp per Android and iOS guidelines.	High
FR-10	The app SHALL be fully compatible with Android TalkBack and iOS VoiceOver screen readers.	High
FR-11	The app SHALL display an animated processing indicator with stage labels (Scanning... / Analysing Diagrams... / Generating Explanation... / Preparing Audio...).	High
FR-12	The Q&A section SHALL show a conversation thread (user question → AI answer) in a scrollable chat-style list.	High

3.1.2 Hardware Interfaces [APP]

ID	Requirement	Priority
FR-13	The app SHALL access the device rear camera using Flutter camera plugin (Camera2 API / AVFoundation).	High
FR-14	The app SHALL request microphone permission for voice-based Q&A input (STT).	Medium

ID	Requirement	Priorty
FR-15	The app SHALL output TTS audio through the device speaker and support Bluetooth headphone routing.	High
FR-16	The app MAY use the device flashlight/torch to illuminate the textbook when capturing in low light.	Low

3.1.3 Software / API Interfaces

ID	Requirement	Priorty
FR-17	The mobile app SHALL communicate with the FastAPI backend via HTTPS REST API.	High
FR-18	Image upload SHALL use HTTP POST multipart/form-data to endpoint POST /api/v1/process.	High
FR-19	The backend SHALL return a structured JSON response: { ocr_text, layout, visuals, explanation, audio_url, session_id }.	High
FR-20	Q&A requests SHALL use POST /api/v1/qa with payload { question, session_id, history }.	High
FR-21	Audio SHALL be returned as a pre-signed URL to an MP3/WAV file streamed from server storage.	High
FR-22	The app SHALL fall back to Google MLKit (Android) or Apple Vision (iOS) for on-device OCR when offline.	Medium
FR-23	The app SHALL use on-device Piper TTS for offline audio generation for short explanations.	Medium

3.1.4 Communication Interfaces

- FR-24: All client-server communication SHALL use TLS 1.3 (HTTPS). Plain HTTP SHALL be rejected.
- FR-25: The app SHALL implement request timeout of 30 seconds for API calls, with a user-visible retry prompt on timeout.
- FR-26: API responses exceeding 5 seconds SHALL display a progress message: Still processing your page.
- FR-27: Session tokens SHALL be JWT-based with 24-hour expiry and silent refresh mechanism.

3.2 Functional Requirements by Module

3.2.1 Module 1 – Camera Capture and Image Input [APP]

ID	Requirement	Priority
FR-28	The camera module SHALL display a real-time viewfinder with an auto-detect page boundary overlay (rectangle guide).	High
FR-29	The app SHALL perform automatic image sharpness assessment and prompt the user to retake if blur score < threshold.	Medium
FR-30	The app SHALL support single-tap capture and also a 3-second timer capture mode for hands-free use by visually impaired users.	High
FR-31	After capture, the app SHALL show a preview screen with options: Use This Image / Retake / Crop.	High
FR-32	The crop tool SHALL include an auto-detect page border feature using edge detection.	Medium
FR-33	The app SHALL accept PDF files and automatically render each page as an image for processing.	High
FR-34	Maximum accepted image file size SHALL be 15 MB; PDF maximum SHALL be 50 MB (first 20 pages).	High

3.2.2 Module 2 – Preprocessing

ID	Requirement	Priority
FR-35	The app SHALL perform on-device preprocessing: deskew (up to $\pm 20^\circ$), resize to 300 DPI equivalent, noise reduction.	High
FR-36	Color correction SHALL normalize over/under-exposed textbook images to improve OCR accuracy.	Medium
FR-37	Preprocessing SHALL complete within 2 seconds on a mid-range device (Snapdragon 700-series or equivalent).	High
FR-38	The preprocessed image SHALL be compressed to JPEG quality 85% before upload to reduce bandwidth usage.	High

3.2.3 Module 3 – OCR and Layout Analysis

ID	Requirement	Priority
FR-39	The backend SHALL extract all text using TrOCR (primary) with Tesseract 5.x as fallback.	High
FR-40	The system SHALL identify layout elements: headings, subheadings, paragraphs, tables, captions, footnotes, bullet lists, equations.	High
FR-41	The system SHALL reconstruct correct reading order for multi-column textbook pages.	High
FR-42	The system SHALL extract text embedded inside diagrams and associate it with the parent visual region.	Medium

ID	Requirement	Priority
FR-43	OCR output SHALL include character-level confidence scores; regions with confidence < 60% SHALL be flagged.	Medium
FR-44	Layout segmentation SHALL use LayoutLMv3 or Donut model on the backend server.	High

3.2.4 Module 4 – Vision Understanding (Diagram / Chart / Equation)

ID	Requirement	Priority
FR-45	The system SHALL detect and classify visual regions as: diagram, chart, graph, math equation, table, or photo.	High
FR-46	The system SHALL generate a natural-language description of each diagram using LLaVA v1.5 or equivalent VLM.	High
FR-47	Chart interpretation SHALL identify: chart type, axis labels, data trends, legend items, and key values.	High
FR-48	Mathematical expressions SHALL be parsed and described in natural English (e.g., 'x squared plus 2x minus 5 equals zero').	High
FR-49	The system SHALL identify spatial relationships in diagrams (e.g., 'Arrow A points from the nucleus to the cytoplasm').	Medium
FR-50	Supported diagram types SHALL include: science diagrams, flowcharts, geometry figures, bar/line/pie charts, labelled maps, and circuit diagrams.	High
FR-51	The system SHALL return a confidence score for each visual description; low-confidence descriptions SHALL be marked for user awareness.	Medium

3.2.5 Module 5 – LLM Explanation Generation

ID	Requirement	Priority
FR-52	The LLM SHALL combine OCR text and visual descriptions into a single coherent explanation of the textbook page.	High
FR-53	The explanation SHALL be generated in three modes: Child (8–12), Teen (13–16), Adult.	High
FR-54	Child mode: vocabulary at Class 5 reading level; includes analogies and simple examples; sentences ≤ 15 words.	High
FR-55	Teen mode: standard educational vocabulary; includes technical terms with brief definitions; structured paragraphs.	High
FR-56	Adult mode: full technical content; formal language; includes formulae references where appropriate.	High
FR-57	The LLM SHALL use LLaMA-3-8B-Instruct or Qwen2-72B hosted on the backend server.	High

ID	Requirement	Priorty
FR-58	Explanations SHALL be grounded strictly in the uploaded page content to minimize hallucination.	High
FR-59	Explanation length SHALL be 150–400 words for Child mode, 250–500 words for Teen, 300–600 words for Adult.	Medium
FR-60	The system SHALL detect if the page has no educational content (e.g., blank page, cover page) and inform the user.	Medium

3.2.6 Module 6 – Text-to-Speech (TTS) [APP]

ID	Requirement	Priorty
FR-61	The app SHALL convert the explanation to audio using Coqui TTS (server-side) or Piper TTS (on-device offline).	High
FR-62	The audio player SHALL have: Play/Pause, Rewind 10s, Forward 10s, Speed control (0.75x / 1x / 1.25x / 1.5x / 2x).	High
FR-63	The audio player SHALL highlight the sentence currently being spoken in the text display below (karaoke-style sync).	Medium
FR-64	TTS SHALL correctly pronounce scientific terms and mathematical expressions read aloud.	Medium
FR-65	The generated audio file SHALL be cached locally so the user can replay without re-requesting from the server.	High
FR-66	TTS audio quality SHALL achieve MOS ≥ 3.5 in user evaluation.	High

3.2.7 Module 7 – Interactive Q&A [APP]

ID	Requirement	Priorty
FR-67	After explanation playback, the app SHALL display a Q&A input area (text field + microphone button).	High
FR-68	Voice input via microphone SHALL use on-device STT (Google / Apple) to convert speech to text.	Medium
FR-69	The system SHALL answer questions using the stored context of the current page (session-aware RAG).	High
FR-70	Answers SHALL be displayed as text and read aloud via TTS automatically.	High
FR-71	The Q&A module SHALL support at least 15 consecutive turns without context loss.	Medium
FR-72	The system SHALL handle off-topic questions gracefully with a response like: 'That question doesn't seem to be about this page. Want me to answer anyway?'	Medium
FR-73	Predefined quick question chips SHALL appear below the player: 'Explain Again', 'Simpler Please', 'Give an Example', 'What Does the Diagram Show?'	High

3.2.8 Module 8 – Adaptive Learning and Feedback [APP]

ID	Requirement	Priority
FR-74	At the end of each explanation, the app SHALL display a 3-option feedback prompt: Too Easy / Just Right / Too Hard.	High
FR-75	Feedback SHALL adjust the complexity of the next explanation within the same session.	Medium
FR-76	The app SHALL maintain a session-level difficulty score updated after each feedback interaction.	Medium
FR-77	The app SHALL log all sessions (page capture, explanation, Q&A exchanges, feedback) to local SQLite.	Medium
FR-78	Session data SHALL be synced to the backend when internet is available, for future personalization.	Low

3.2.9 Module 9 – User Profile and History [APP]

ID	Requirement	Priority
FR-79	During first launch, the app SHALL run an onboarding wizard: Name → Age Level → Accessibility Preferences → Notification Preferences.	High
FR-80	The app SHALL store user profile on-device (encrypted) and optionally sync to backend.	Medium
FR-81	The History tab SHALL list all past sessions sorted by date with: thumbnail, date, subject tag, and duration.	Medium
FR-82	The user SHALL be able to tap any history item to replay the explanation audio and re-enter Q&A.	Medium
FR-83	The app SHALL allow users to bookmark/save specific explanations for quick access.	Low
FR-84	The user SHALL be able to delete individual history items or clear all history from Settings.	Medium

3.3 Non-Functional Requirements

3.3.1 Performance

ID	Requirement	Target Metric
NFR-01	End-to-end pipeline: camera capture to first audio output.	≤ 25 seconds on 4G; ≤ 15 seconds on Wi-Fi.
NFR-02	On-device preprocessing time.	≤ 2 seconds on mid-range Android (Snapdragon 720G+).

ID	Requirement	Target Metric
NFR-03	OCR Character Error Rate on clean textbook images.	CER < 5%.
NFR-04	Explanation quality – BLEU score.	BLEU $\geq$ 0.35 vs human reference.
NFR-05	TTS Mean Opinion Score.	MOS $\geq$ 3.5 / 5.0.
NFR-06	App launch time (cold start).	$\leq$ 3 seconds on supported devices.
NFR-07	App battery drain per 30-minute session.	$\leq$ 10% on a 4000 mAh battery.
NFR-08	API server support for concurrent users.	$\geq$ 20 concurrent users without >50% latency increase.

3.3.2 Safety and Content

- NFR-09: The LLM output SHALL pass a content safety filter before display. No harmful, offensive, or age-inappropriate content SHALL reach the user.
- NFR-10: Uploaded images SHALL NOT be retained on the server beyond 48 hours. Auto-deletion policy MUST be enforced.
- NFR-11: The app SHALL NOT silently fail. Any processing error SHALL display a clear, user-friendly error message with recovery options.
- NFR-12: If OCR confidence is critically low, the app SHALL warn the user: 'Image quality is poor. Results may be inaccurate. Please retake with better lighting.'

3.3.3 Security [APP]

- NFR-13: All server communication SHALL use TLS 1.3. HTTP traffic SHALL be blocked via network security config.
- NFR-14: User credentials and session tokens SHALL be stored using Flutter Secure Storage (Android Keystore / iOS Keychain).
- NFR-15: The app SHALL implement certificate pinning for the backend API domain to prevent man-in-the-middle attacks.
- NFR-16: Uploaded images SHALL be transmitted as encrypted multipart requests; no plain-text image bytes.
- NFR-17: Admin backend access SHALL require two-factor authentication.

3.3.4 Reliability and Availability

- NFR-18: Backend server uptime SHALL be $\geq$ 95% during evaluation and demo periods.
- NFR-19: Module failure isolation: if the VLM diagram module fails, the system SHALL proceed with OCR-only explanation and notify the user.
- NFR-20: Offline mode (OCR + TTS) SHALL be available 100% of the time regardless of internet connectivity.

- NFR-21: The app SHALL implement automatic retry (up to 3 attempts) for failed API calls with exponential backoff.

3.3.5 Usability and Accessibility [APP]

- NFR-22: A new user with no prior AI app experience SHALL complete first scan and hear explanation within 3 minutes of installing the app.
- NFR-23: All text on screen SHALL be readable at system large font size without truncation or overflow.
- NFR-24: A visually impaired user relying solely on TalkBack/VoiceOver SHALL be able to complete the full workflow without sighted assistance.
- NFR-25: The app SHALL follow WCAG 2.1 Level AA guidelines adapted for mobile (WCAG Mobile Extension, W3C).
- NFR-26: Color SHALL NOT be the only means of conveying information (e.g., error states use icons + text, not color alone).

3.3.6 Maintainability and Scalability

- NFR-27: The Flutter frontend and FastAPI backend SHALL be fully decoupled; the AI models on the backend SHALL be replaceable without modifying the app.
- NFR-28: The backend SHALL be containerized with Docker; each AI module SHALL run in its own container.
- NFR-29: The app SHALL be structured using Clean Architecture (Presentation / Domain / Data layers) for maintainability.
- NFR-30: The codebase SHALL include unit tests covering $\geq 60\%$ of business logic (Flutter + Python).

3.3.7 Portability [APP]

- NFR-31: The Flutter app SHALL build and run on Android 10+ (API 29+) and iOS 14+ from a single codebase with platform-specific adapters for camera, STT, and secure storage.
- NFR-32: The app APK/IPA SHALL not exceed 100 MB for initial install; AI model assets SHALL be downloaded on first launch.

4. System Architecture – Mobile Application

4.1 High-Level Architecture

MATA follows a two-tier architecture: a Flutter mobile client (Tier 1) and a FastAPI AI backend server (Tier 2). Heavy AI computation (VLM, LLM) is offloaded to the server; lightweight operations (preprocessing, on-device OCR, TTS) run locally on the device.

Tier	Component	Technology	Responsibility
Mobile Client (T1)	UI Layer	Flutter / Dart	Camera, navigation, audio player, Q&A UI
Mobile Client (T1)	Domain Layer	Flutter BLoC / Riverpod	Business logic, state management, offline logic
Mobile Client (T1)	Data Layer	Dio HTTP, SQLite, Hive	API calls, local storage, caching
Mobile Client (T1)	On-Device AI	TFLite / ONNX Runtime	Lightweight OCR (MLKit), offline TTS (Piper)
Backend Server (T2)	API Gateway	FastAPI + NGINX	Request routing, auth, rate limiting
Backend Server (T2)	OCR Service	TrOCR + Tesseract	Text extraction and layout analysis
Backend Server (T2)	Vision Service	LLaVA v1.5 + DETR	Diagram/chart description, math parsing
Backend Server (T2)	LLM Service	LLaMA-3-8B / Qwen2-72B	Explanation generation and Q&A reasoning
Backend Server (T2)	TTS Service	Coqui TTS / Tacotron 2	Server-side neural speech synthesis
Backend Server (T2)	Data Store	PostgreSQL + Redis	User profiles, sessions, model cache

4.2 App Screen Map (Navigation Structure)

Screen	Description	Key Components
Splash Screen	Brand animation on app launch.	Logo, loading indicator.
Onboarding (1–3)	First-time setup: name, age level, accessibility prefs.	Page indicators, selectors, CTA button.
Home / Scan	Camera viewfinder – main entry point.	Camera preview, capture button, gallery/PDF import.
Preview / Crop	Review captured image before processing.	Image preview, crop handles, Confirm / Retake buttons.
Processing	Pipeline animation while AI works.	Stage labels, animated progress, cancel option.

Screen	Description	Key Components
Result	Explanation text + audio player + Q&A.	Text scroll view, audio bar, Q&A thread, feedback chips.
History	List of past sessions.	Date-sorted list, thumbnails, subject tags, delete option.
History Detail	Replay any saved session.	Explanation text, audio player, re-enter Q&A.
Profile	User settings and preferences.	Name, age level, accessibility, notification toggles.
Settings	App configuration.	Dark/light mode, font size, language, data management.

4.3 API Endpoint Specification

Meth od	Endpoint	Description	Request	Response
POST	/api/v1/auth/register	Register new user.	{ name, email, age_level }	{ user_id, token }
POST	/api/v1/auth/login	Login existing user.	{ email, password }	{ token, expiry }
POST	/api/v1/process	Process textbook page image.	multipart: image_file, age_level, uid	{ session_id, ocr, explanation, audio_url }
POST	/api/v1/qa	Q&A on current session.	{ session_id, question, history }	{ answer, audio_url }
POST	/api/v1/feedback	Submit difficulty feedback.	{ session_id, rating }	{ ack, next_level }
GET	/api/v1/history	Get all user sessions.	Header: Authorization: Bearer <token>	{ sessions[] }
GET	/api/v1/health	Backend health check.	None	{ status, gpu_available }

5. Other Requirements

5.1 Hardware and Software Summary

Layer	Component	Technology / Version
Mobile App	Framework	Flutter 3.x (Dart 3.x)
Mobile App	State Management	Riverpod 2.x or BLoC
Mobile App	HTTP Client	Dio 5.x
Mobile App	Local Database	SQLite via sqflite + Hive
Mobile App	Secure Storage	flutter_secure_storage
Mobile App	Camera	camera plugin (Camera2 / AVFoundation)
Mobile App	STT (Voice Input)	speech_to_text plugin (Google / Apple)
Mobile App	On-device TTS	flutter_tts plugin + Piper model
Mobile App	On-device OCR	google_mlkit_text_recognition / Apple Vision
Mobile App	PDF Rendering	pdf_render / pdfx Flutter plugin
Backend	API Framework	FastAPI 0.100+ (Python 3.10+)
Backend	OCR	TrOCR (Hugging Face) + Tesseract 5.x
Backend	Layout	LayoutLMv3 / Donut (Hugging Face)
Backend	Vision / VLM	LLaVA v1.5 (7B) via Hugging Face
Backend	Math Parsing	EDSL / pix2tex / lm2LaTeX
Backend	LLM	LLaMA-3-8B-Instruct (Meta) or Qwen2-72B
Backend	TTS	Coqui TTS (XTTS v2) or Tacotron 2
Backend	Database	PostgreSQL 14+ / Redis 7+
Backend	Deployment	Docker + NGINX + Ubuntu 22.04

5.2 Evaluation Metrics

Module	Metric	Target
OCR	Character Error Rate (CER)	< 5% on clean images
OCR	Word Error Rate (WER)	< 10%
Explanation	BLEU Score	≥ 0.35
Explanation	ROUGE-L Score	≥ 0.40
Explanation	Human rating (1–5 scale)	≥ 3.5 / 5.0
Diagram Desc.	Human accuracy rating	≥ 70%
TTS	Mean Opinion Score (MOS)	≥ 3.5 / 5.0

Module	Metric	Target
Performance	E2E pipeline time (Wi-Fi, GPU backend)	≤ 15 seconds
Performance	E2E pipeline time (4G, GPU backend)	≤ 25 seconds
App Stability	Crash-free rate (Firebase Crashlytics)	$\geq 98\%$
Battery	Drain per 30-min session	$\leq 10\%$ on 4000 mAh

5.3 Legal and Compliance

- NFR-33: The app SHALL comply with India's Digital Personal Data Protection Act 2023 (DPDPA) for handling user data.
- NFR-34: All pre-trained AI models SHALL be used in compliance with their licenses: Meta LLAMA Community License, Apache 2.0, MIT, Mozilla Public License.
- NFR-35: The app MUST display a Privacy Policy and Terms of Use during onboarding, requiring explicit user consent.
- NFR-36: User-uploaded textbook images SHALL NOT be used for model training without explicit written consent from the user.
- NFR-37: The app SHALL comply with Google Play Developer Policy and Apple App Store Review Guidelines.
- NFR-38: The app SHALL include a COPPA-compliant consent mechanism for users under age 13 (parental consent required).

6. Unique Market Differentiators and Innovation Ideas

About This Section

This section proposes 8 unique feature ideas that go beyond the core MATA concept. Each idea is grounded in real market gaps, designed to make MATA more competitive, commercially viable, and impactful. These are roadmap features for v2.0 and beyond.

6.1 Idea 1 – 'Scan-and-Quiz' Mode (Instant Test Generation)

Attribute	Detail
Core Idea	After generating an explanation, MATA automatically generates 3–5 quiz questions from the textbook page content. The student answers them via voice or tap; the app grades and explains wrong answers.

Attribute	Detail
Why Unique	No accessibility-focused app currently combines explanation AND self-testing in a single camera scan. Existing quiz apps (Quizlet, Kahoot) require manual content entry.
Market Gap	Students lack immediate self-assessment tools tied to the exact textbook material they are studying at that moment.
Tech Required	LLM prompt engineering for question generation (MCQ / True-False / Fill-in-the-blank). On-device answer evaluation.
User Benefit	Converts passive listening to active learning. Proven to improve retention by 30–40% (spacing effect research).
Monetization	Premium feature in MATA Pro subscription.

6.2 Idea 2 – 'Subject Tagging + Smart Study Planner'

Attribute	Detail
Core Idea	MATA automatically detects the subject of each scanned page (Physics, Chemistry, Math, History, Biology etc.) using the LLM. It then builds a personal study timeline showing how many pages the student has covered per subject and suggests what to study next.
Why Unique	Combines an AI textbook reader with a lightweight study planner — no app currently does this from physical textbook scanning.
Market Gap	Students (especially Class 10–12 board exam students) struggle to plan revision. A tool that tracks what they've actually read (not just what they plan to read) is unprecedented.
Tech Required	LLM-based subject and chapter classification. Local SQLite history + visualization charts in Flutter.
User Benefit	Reduces exam anxiety; provides a concrete picture of revision coverage.
Monetization	Freemium: basic subject detection free; full study planner + progress analytics in MATA Pro.

6.3 Idea 3 – Vernacular Language Mode (Bhasha Bridge)

Attribute	Detail
Core Idea	After generating an explanation in English, MATA offers a 'Translate & Read' button that translates the explanation into 10+ Indian regional languages (Hindi, Marathi, Tamil, Telugu, Bengali, Kannada, Gujarati, Malayalam, Punjabi, Odia) and reads it aloud in that language.
Why Unique	No existing accessible textbook tool supports regional Indian language audio explanations from English textbooks. This directly targets India's 200M+ students studying in regional-medium schools.
Market Gap	NCERT and state board textbooks are in English but millions of students think and comprehend better in their mother tongue.
Tech Required	IndicTrans2 (AI4Bharat) for Indian language translation. MMS-TTS (Meta) or IndicTTS for multilingual speech output.
User Benefit	Bridges the language gap for Tier 2/3 city students. Huge social impact potential.

Attribute	Detail
Monetization	Strong CSR / EdTech grant potential. Premium language packs.

6.4 Idea 4 – 'Concept Map' Visual Generator

Attribute	Detail
Core Idea	After processing a page, MATA generates an interactive concept map (mind map) showing the key concepts on the page and their relationships. The map is displayed visually in the app and can be exported as an image.
Why Unique	Traditional mind-map tools (XMind, MindNode) require manual input. MATA auto-generates a concept map from a single camera scan of a textbook page — entirely automated.
Market Gap	Visual learners and students who need structured note-taking benefit enormously from auto-generated concept maps, but no tool offers this from a camera scan.
Tech Required	LLM extracts key concepts + relationships → graph-based rendering in Flutter (graphview package or custom Canvas widget). Export to PNG.
User Benefit	One-tap visual revision notes. Drastically cuts study time.
Monetization	Shareable concept maps drive viral growth. Premium: export to PDF, cloud sync, collaborative maps.

6.5 Idea 5 – 'Teacher Mode' and Classroom Broadcast

Attribute	Detail
Core Idea	A teacher scans a textbook page on their device; MATA's explanation is instantly broadcast (via QR code or a class session code) to all students' phones in the classroom. Students can listen, ask questions, and vote on difficulty — all in real-time.
Why Unique	Transforms MATA from a personal tool into a classroom collaboration platform. No existing EdTech app combines real-time teacher broadcast with multimodal textbook understanding.
Market Gap	Teachers in Indian schools lack affordable, real-time interactive tools that work with physical textbooks they already use.
Tech Required	WebSocket-based broadcast (FastAPI WebSocket), Flutter session-join via QR scan (camera → QR decode), real-time difficulty voting UI.
User Benefit	Makes MATA institutionally adoptable, not just a consumer app. Enables B2B school licensing model.
Monetization	MATA for Schools: institutional subscription (₹X per school per year). High LTV.

6.6 Idea 6 – Exam Paper Solver and Step Explainer

Attribute	Detail
Core Idea	The student scans a previous year exam question paper or a worksheet. MATA detects each question, solves it step-by-step (for math/science), and reads the solution aloud with explanation at the selected age level.
Why Unique	Combines OCR of question papers + LLM step-by-step reasoning + TTS — no app does all three from a camera scan. Photomath solves math but cannot explain diagrams or non-math questions, and has no TTS.
Market Gap	Board exam students (Class 10/12) desperately need step-by-step explained solutions to past papers with audio support.
Tech Required	Question detection (layout analysis to separate numbered items). LLM chain-of-thought prompting for step-by-step solutions. Separate TTS pass per step.
User Benefit	One of the highest-demand features for exam preparation in India. Massive retention driver.
Monetization	MATA Exam Prep: a separate premium mode. Partnering with exam prep companies (Byju's, Unacademy style licensing).

6.7 Idea 7 – Haptic Diagram Explorer (Accessibility Innovation)

Attribute	Detail
Core Idea	For visually impaired users, MATA provides a 'Haptic Diagram Mode': the user places a finger anywhere on the screen (which mirrors the camera capture of the diagram) and the app vibrates in patterns to indicate edges, labels, and regions. Audio announces what is at the touched region ('You are touching the nucleus → The nucleus controls cell activity').
Why Unique	No existing mobile app provides haptic + audio diagram exploration. Current accessible apps only describe diagrams in audio — they cannot let a blind user interactively explore diagram regions.
Market Gap	Blind and visually impaired students have NO way to interact with diagrams in any existing tool.
Tech Required	Diagram region segmentation (DETR bounding boxes) mapped to screen coordinates. Flutter haptic feedback API (HapticFeedback). Touch region → nearest bounding box → audio label.
User Benefit	Breakthrough accessibility feature. Could win accessibility awards and government recognition.
Monetization	MATA Accessibility Edition — eligible for government grants (NIEPVD, DAISY Consortium), NGO partnerships, NASSCOM Foundation.

6.8 Idea 8 – 'Learning Streak' Gamification + Parental Dashboard

Attribute	Detail
Core Idea	MATA tracks daily learning streaks (pages scanned, questions asked, concepts mastered). Students earn badges and level up. A separate Parental Dashboard app/web view lets parents see their child's study activity, subjects covered, weak areas, and Q&A history.

Attribute	Detail
Why Unique	No accessibility-focused educational app combines gamification + parental visibility + AI textbook reading. Duolingo's streak model applied to textbook learning.
Market Gap	Parents of visually impaired children have no visibility into what and how much their child is independently studying.
Tech Required	Flutter gamification widgets, badge system (local SQLite), push notifications (FCM/APNS), parent web portal (React.js) consuming same backend API.
User Benefit	Drives daily active usage (DAU). Parental dashboard makes MATA a family product, not just a student tool — increasing purchase decision power.
Monetization	Family Plan subscription. Parental dashboard as premium feature.

6.9 Market Positioning Summary

Feature / Idea	Competitor Gap Addressed	Primary Audience	Revenue Model
Core MATA (OCR + VLM + TTS)	No app reads + explains textbook visuals with audio	Blind students, Young learners	Freemium / Pro subscription
Scan-and-Quiz Mode	Quizlet needs manual input; no camera-based quizzing	All students	MATA Pro
Study Planner Integration	No tracker tied to actual physical textbook reading	Class 10–12 students	Freemium Analytics
Vernacular Language Mode	No Indian-language textbook audio explainer	Tier 2/3 city India	Language pack IAP / Grants
Concept Map Generator	No auto-generated maps from camera scan	Visual learners	MATA Pro export feature
Teacher Mode / Broadcast	No real-time textbook-scan classroom tool	Schools, Teachers	B2B school license
Exam Paper Solver	Photomath = only math; no audio; no diagrams	Board exam students	MATA Exam Prep tier
Haptic Diagram Explorer	Zero existing haptic diagram tools for blind users	Visually impaired	Grants + Accessibility Edition
Gamification + Parent Dashboard	No parent visibility in accessibility EdTech	Families, Parents	Family Plan

7. Appendices

Appendix A: Requirements Traceability Matrix (RTM)

FR Range	Module	Flutter Layer	Backend Component	Test Method
FR-01 to FR-12	App UI / Navigation	Presentation Layer	—	UI Test / WCAG Audit
FR-13 to FR-16	Camera & Hardware	Platform Channel	—	Device Integration Test
FR-17 to FR-27	API Communication	Data Layer / Dio	FastAPI Gateway	API Contract Test
FR-28 to FR-34	Camera Capture & Import	Presentation + Domain	—	Functional Test
FR-35 to FR-38	Preprocessing	Domain + Data	OpenCV / PIL service	CER Evaluation
FR-39 to FR-44	OCR + Layout	Data Layer (API call)	TrOCR + LayoutLMv3	CER / WER Test
FR-45 to FR-51	Vision Understanding	Data Layer (API call)	LLaVA + DETR	Human Accuracy Rating
FR-52 to FR-60	LLM Explanation	Data Layer (API call)	LLaMA / Qwen2	BLEU / ROUGE / Human
FR-61 to FR-66	TTS + Audio Player	Presentation + Data	Coqui TTS / Piper	MOS Evaluation
FR-67 to FR-73	Interactive Q&A	Presentation + Domain	LLM + Context Store	QA Relevance Test
FR-74 to FR-78	Adaptive Learning	Domain + Data	PostgreSQL + LLM prompt	Behavioral Observation
FR-79 to FR-84	Profile + History	Data Layer / SQLite	PostgreSQL	Functional Test

Appendix B: Technology Stack Summary

Category	Technology	Version	License
Mobile Framework	Flutter + Dart	3.x / 3.x	BSD 3-Clause (Google)
State Management	Riverpod	2.x	MIT
HTTP Client	Dio	5.x	MIT
Local DB	sqlite + Hive	Latest	MIT
Secure Storage	flutter_secure_storage	Latest	BSD
Camera	camera	Latest	BSD (Flutter)
On-device OCR	google_mlkit_text_rec.	Latest	Apache 2.0

Category	Technology	Version	License
On-device TTS	flutter_tts + Piper	Latest	MIT / Mozilla
PDF Rendering	pdfx	Latest	Apache 2.0
Backend Framework	FastAPI + Uvicorn	0.100+ / 0.18+	MIT
OCR (Cloud)	TrOCR (Hugging Face)	Latest	MIT
Layout Analysis	LayoutLMv3	Latest	CC BY-NC 4.0
VLM (Vision)	LLaVA v1.5	7B params	Apache 2.0
LLM	LLaMA-3-8B-Instruct	8B params	Meta LLAMA License
TTS (Cloud)	Coqui XTTS v2	Latest	Mozilla PL 2.0
Math Parsing	pix2tex / EDSL	Latest	MIT / Apache
Translation (Idea 3)	IndicTrans2 (AI4Bharat)	Latest	MIT
Database	PostgreSQL + Redis	14+ / 7+	PostgreSQL License / BSD
Deployment	Docker + NGINX	20+ / Latest	Apache 2.0

Appendix C: Dataset References

Dataset	Purpose	Source
PubLayNet / DocLayNet	Document layout segmentation training & eval	IBM Research
CROHME	Mathematical expression recognition benchmark	ICDAR
Im2LaTeX-100K	Equation to LaTeX conversion	Academic (OpenAI)
AI2D (Allen Institute)	Science diagram understanding benchmark	Allen AI
FUNSD	Form understanding benchmark	Research release
IIT-CDIP	Document image OCR training corpus	Illinois Institute of Tech.
Custom MITAOE Dataset	Internal textbook page collection for testing	Collected by project team
Bhasha Drishti / IndicOCR	Indic language OCR training (for Idea 3)	AI4Bharat / CDAC

End of Software Requirements Specification

MATA – Multimodal Accessible Textbook Assistant (Mobile Application Edition)

Version 2.0 | MIT Academy of Engineering, Pune | Academic Year 2025–2026